



1. What is the value of $\frac{21^2 + \frac{21}{20}}{20^2 + \frac{20}{21}}$?

- (A) $\frac{400}{441}$ (B) $\frac{20}{21}$ (C) 1 (D) $\frac{21}{20}$ (E) $\frac{441}{400}$

2. Two circles of equal radius r have an overlap of area 7π and the total area covered by the circles is 25π . What is the value of r ?

- (A) $2\sqrt{2}$ (B) $2\sqrt{3}$ (C) 4 (D) 5 (E) $4\sqrt{2}$

3. A pyramid whose base is a regular n -gon has the same number of edges as a prism whose base is a regular m -gon. What is the smallest possible value of n ?

- (A) 3 (B) 4 (C) 6 (D) 9 (E) 12

4. There exists a positive integer N such that $\frac{\frac{1}{999} + \frac{1}{1001}}{2} = \frac{1}{1000} + \frac{1}{N}$. What is the sum of the digits of N ?

- (A) 36 (B) 45 (C) 54 (D) 63 (E) 72

5. For positive integers $m > 1$ and $n > 1$, the sum of the first m multiples of n is 2020. Compute $m + n$.

- (A) 206 (B) 208 (C) 210 (D) 212 (E) 214

6. How many of the following statements are true for every parallelogram $\mathcal{P}$?
The perpendicular bisectors of the sides of $\mathcal{P}$ share at least one common point.
The perpendicular bisectors of the sides of $\mathcal{P}$ are all distinct.

If the perpendicular bisectors of the sides of $\mathcal{P}$ share at least one common point then $\mathcal{P}$ is a square.

If the perpendicular bisectors of the sides of $\mathcal{P}$ are all distinct then these bisectors form a parallelogram.

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

7. It is known that every positive integer can be represented as the sum of at most 4 squares. What is the sum of the 2 smallest integers which cannot be represented as a sum of fewer than 4 squares?

(A) 22 (B) 23 (C) 24 (D) 25 (E) 26

8. There are 8 scoops of ice cream, two of each flavor: vanilla, strawberry, chocolate, and mint. If scoops of the same flavor are not distinguishable, how many ways are there to distribute one scoop of ice cream to each of 5 different people?

(A) 420 (B) 450 (C) 600 (D) 1620 (E) 2520

9. Suppose points A, B, C, D, E, F lie on a line such that $AB = 1$, $BC = 4$, $CD = 9$, $DE = 16$, $EF = 25$, and $FA = 23$. What is CF ?

(A) 14 (B) 18 (C) 21 (D) 25 (E) 26

10. A soccer league consists of 20 teams and every pair of teams play each other once. If the game is a draw then each team receives one point, otherwise the winner receives 3 points while the loser receives 0 points. If the total number of points scored by all the teams was 500, how many games ended in a draw?

(A) 70 (B) 75 (C) 100 (D) 115 (E) 120

11. Suppose that $ABCD$ is a parallelogram whose diagonals intersect at P . Let X and Y be the feet from P to AD and AB , respectively. If $AX = 1$, $AY = 2$, and $BY = 5$, then compute DX .

(A) $\sqrt{22}$ (B) $2 + 2\sqrt{2}$ (C) $2\sqrt{6}$ (D) $\frac{11}{2}$ (E) 6

12. Let a, b be not necessarily distinct positive integers chosen uniformly at random from the set $\{1, 2, \dots, 100\}$. Find the probability that $\lfloor \sqrt{a} \rfloor + \lceil \sqrt{b} \rceil = \lceil \sqrt{a} \rceil + \lfloor \sqrt{b} \rfloor$.

- (A) $\frac{17}{25}$ (B) $\frac{39}{50}$ (C) $\frac{81}{100}$ (D) $\frac{41}{50}$ (E) $\frac{21}{25}$

13. Jerry writes an equation on a board. However, Mario comes along and erases some of the digits, and replaces them with variables. On the board remains $\log_a \overline{b2c} = d$ for $a \neq 1$ and $b \neq 0$. Find the sum of all possible values of d .

- (A) 13 (B) 17 (C) 18 (D) 20 (E) 24

14. Let a and b be real numbers such that $z_1 = 1010 + 2021i$, $z_2 = 2020 + 1011i$ and $z_3 = az_1 + bz_2$ are the vertices of an equilateral triangle in the complex plane. What is the sum of all possible values of $a + b$?

- (A) -6062 (B) -3031 (C) 2 (D) 3031 (E) 6062

15. For some reals a, b, c, d , a polynomial $P(x) = x^3 + ax^2 + bx + 10$ has three distinct real roots in an arithmetic progression, with each root precisely 1 less than a root of $Q(x) = x^3 - 7x^2 + cx + d$. If b can be written as $-\frac{m}{n}$ for relatively prime positive integers m and n , compute $m + n$.

- (A) 88 (B) 89 (C) 90 (D) 91 (E) 92

16. Let $ABCD$ be a quadrilateral inscribed in a circle ω . Diagonals AC and BD meet at P such that $BP = 4$ and $DP = 7$. Point Q is chosen such that $\triangle CPQ$ is equilateral. What is the area of $\triangle APQ$?

- (A) 9 (B) 12 (C) $7\sqrt{3}$ (D) $14 - \sqrt{3}$ (E) $12 + \sqrt{3}$

17. We say that two quadratic polynomials $P(x)$ and $Q(x)$ with leading coefficient 1 are *friends* if for every root r of $P(x)$, $Q(r)$ is also a root of $P(x)$. If $P(x) = x^2 - 4x + 1$, what is the maximum value of $Q(0)$ over all friends of $P(x)$?

- (A) 1 (B) $3 - \sqrt{3}$ (C) $3 + \sqrt{3}$ (D) 5 (E) 6

18. Pyramid $ABCDE$ with square base $BCDE$ and apex A has edge lengths all equal to 1 with the square base sitting atop a table. Some of the edges of the pyramid are cut so that it is opened out flat to its net (in one piece) on the table without any folds. Two possibilities for the net are shown below, and nets are considered the same if they differ by a rotation but different if they differ by a reflection. How many distinct nets are possible?

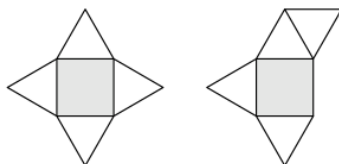


Figure 1: Problem 18

- (A) 10 (B) 11 (C) 13 (D) 15 (E) 16

19. Consider the function $f(x) = \underbrace{2 \sin(2 \sin(2 \sin(\dots 2 \sin(2 \sin(x)) \dots)))}_{200 \sin's}$. Over the interval $0 \leq x \leq \pi$, $f(x)$ attains its maximum value M a total of N times. Determine the largest integer no greater than $M + N$.

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

20. For a certain positive integer n , there are exactly 2021 ordered pairs of positive divisors (d_1, d_2) of n for which d_1 and d_2 are relatively prime. What is the sum of all possible values of the number of divisors of n ?

- (A) 1535 (B) 1536 (C) 1537 (D) 1538 (E) 1539

21. Circles ω and γ have radii 2 and 3 respectively, and are externally tangent at A . Let $ABCDEF$ be a non-degenerate regular hexagon such that B lies on ω and E lies on γ . The area of this hexagon can be written as $\frac{a\sqrt{b}}{c}$ where a, b, c are positive integers with a, c coprime and b square free. Find $a + b + c$.

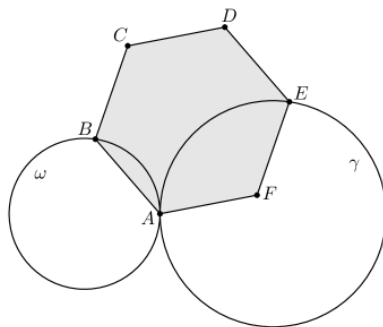


Figure 2: Problem 21

(A) 80 (B) 81 (C) 82 (D) 83 (E) 84

22. Find the number of integers n between 2 and 50, inclusive, such that the product of any two distinct divisors of n is either a multiple or divisor of n .

(A) 21 (B) 25 (C) 28 (D) 33 (E) 36

23. How many ordered tuples of positive integers $(a_1, a_2, \dots, a_n)$ are there such that $\sum_{k=1}^n a_k = 12$ and $\sum_{k=1}^{\ell} (-1)^k a_k \leq \sum_{k=1}^n (-1)^k a_k = 0$ for all $\ell \in \{1, 2, \dots, n-1\}$?

(A) 64 (B) 128 (C) 132 (D) 144 (E) 256

24. In triangle $\triangle ABC$, $AB = 5$, $BC = 6$, and $CA = 7$. Let ω be the circle with diameter BC , and let $\overline{AE}$ and $\overline{AF}$ be the tangents from A to ω , such that segments BE and CF intersect at a point P inside $\triangle ABC$. What is AP ?

(A) 3 (B) $\sqrt{19}$ (C) $\sqrt{21}$ (D) $2\sqrt{7}$ (E) 6

25. In a set of 15 points, no 4 of which are cyclic and no 3 of which are collinear, there are exactly 2021 pairs $(\mathcal{C}, P)$ where $\mathcal{C}$ is a circle that passes through 3 of the points and P is a point lying strictly inside $\mathcal{C}$. What is the maximum number of convex quadrilaterals with vertices among the 15 points?

(A) 637 (B) 656 (C) 678 (D) 715 (E) 734

Time limit: 75 minutes